TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

Bachelor of Artificial Intelligence and Robotics Engineering

AI.405

Robot Manipulation

COURSE OVERVIEW

Robot grasping and manipulation: grippers, grasp planning, dexterous manipulation, and contact.

ROBOT MANIPULATION

1

SESSION

60 MINUTES

ORIENTATION: GRASPING THE WORLD

KEY TOPICS

- Manipulation in robotics
- Grasp types
- Contact mechanics
- Applications

TEACHING FOCUS

Let robots hold and move objects.

KEY TAKEAWAY Manipulation is physical intelligence.

2

SESSION

60 MINUTES

GRIPPERS & END-EFFECTORS

KEY TOPICS

- Parallel-jaw grippers
- Vacuum and adhesive
- Multi-fingered hands
- End-effector selection

TEACHING FOCUS

Choose the right hand for the task.

▲ **KEY TAKEAWAY** The gripper is the robot's hand.

ROBOT MANIPULATION

3

SESSION

60 MINUTES

GRASP PLANNING

KEY TOPICS

- Grasp taxonomy
- Force-closure and form-closure
- Grasp quality metrics
- Grasp synthesis

TEACHING FOCUS

Plan stable and effective grasps.

KEY TAKEAWAY A good grasp resists disturbance.

4

SESSION

60 MINUTES

DEXTEROUS MANIPULATION

KEY TOPICS

- In-hand manipulation
- Finger gaiting
- Tactile feedback
- Dexterous control

TEACHING FOCUS

Manipulate objects skillfully.

KEY TAKEAWAY Dexterity turns grasping into tool use.

5

SESSION

60 MINUTES

CONTACT & INTERACTION

KEY TOPICS

- Contact modeling
- Impedance control
- Slip detection
- Safe physical interaction

TEACHING FOCUS

Manage contact safely.

KEY TAKEAWAY Contact is where manipulation happens.

ROBOT MANIPULATION

6

SESSION

60 MINUTES

ROBOT MANIPULATION & REVIEW

KEY TOPICS

- Manipulation demo
- Course recap and assessment
- Toward robot perception (AI.406)
- Grasp challenge

TEACHING FOCUS

Pick, place, and manipulate reliably.

KEY TAKEAWAY Manipulation is the robot's handiwork.